FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE — GRADE 10 | SUB-STRAND 3.1: TURNING EFFECT OF FORCE	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS
This is your Final Explanation task. You will use everything you have learned over the past 8 lessons to answer the driving question fully and scientifically. Read the driving question carefully, then write a response that covers each section below. Use correct scientific vocabulary, refer back to evidence you collected, and connect your answer to real Kenyan tools and contexts. A strong response does not just state facts — it explains the science behind what you observed and links ideas together logically. Write in full sentences and use diagrams where they help your explanation.

Section 1: Explain the Anchoring Phenomenon	
Return to the Jembe Handle Puzzle from the start of this unit. A farmer in Kericho can break hard soil easily with a long-handled jembe, but struggles with a short-handled jembe of the same blade weight. Using the scientific concepts you have learned, explain exactly WHY this happens. Your explanation must include the terms: torque (turning effect), pivot, effort force, and perpendicular distance.	When the farmer uses a long-handled jembe, they create a large turning effect — called torque — on the blade at the pivot point (where the blade meets the soil). Torque is calculated as: Torque = Force × Perpendicular distance. Even though the farmer applies the same effort force, the longer handle increases the perpendicular distance from the pivot to the point where the force is applied. A greater perpendicular distance means a greater torque, so the blade turns into the soil more powerfully. When the farmer grips close to the blade, the perpendicular distance is very small, so the torque is small — and the blade barely cuts into the soil even though the same person is pushing just as hard. This is why handle length, not just effort, determines how easy or hard it is to make the jembe turn.

Section 2: The Science of Torque — Explaining the Principle	
Define torque (turning effect of	Torque is the measure of how effectively a force causes an object to rotate about a pivot point. The formula is: Torque (τ) = Force

a force) in your own words. Write the formula and explain what each term means. Then describe how changing either the size of the force OR the distance from the pivot affects the turning effect. Use a worked numerical example — you may invent realistic numbers for a Kenyan farming or everyday tool.	(F) × Perpendicular distance (d), where Force is the push or pull applied (measured in Newtons, N) and perpendicular distance is the shortest distance from the pivot to the line of action of the force (measured in metres, m). Torque is measured in Newton-metres (N·m). For example, imagine Wanjiku applies a 20 N downward force on a jembe handle. If her hands are 0.8 m from the pivot (blade tip), the torque is 20 N × 0.8 m = 16 N·m. If she slides her grip to only 0.2 m from the pivot, the torque drops to 20 N × 0.2 m = 4 N·m — just one quarter as much. This shows that doubling the distance doubles the torque, and halving the distance halves the torque, even when the applied force stays exactly the same.

Section 3: The Principle of Moments and Equilibrium	
Explain the Principle of Moments. What must be true about the clockwise and anticlockwise moments acting on an object for it to be balanced (in equilibrium)? Describe one real Kenyan example — such as a see-saw at a school in Kisumu, a balance scale used in a Nairobi market, or a door — where this principle is clearly at work. Show how the mathematics confirms the balance.	The Principle of Moments states that for an object to be in rotational equilibrium (balanced and not rotating), the total clockwise moment about any pivot must equal the total anticlockwise moment about the same pivot. In other words: Sum of clockwise moments = Sum of anticlockwise moments. Consider a market weighing balance in Nairobi: a trader places a 2 kg bag of maize flour 0.3 m to the left of the pivot, creating an anticlockwise moment of (2 × 10) × 0.3 = 6 N·m. To balance this, the trader places a 500 g (0.5 kg) weight on the right side. For balance, the weight must be placed at a distance d, where (0.5 × 10) × d = 6 N·m, so d = 6 ÷ 5 = 1.2 m from the pivot. This example shows that a smaller force can balance a larger one — as long as it is placed farther from the pivot, consistent with the Principle of Moments.

Section 4: Engineering Applications — How Kenyan Engineers and Farmers Design Better Tools	
Choose at least TWO examples of tools or machines used in Kenya — one from farming and one from another context (construction, transport, the home, or a school workshop). For each example, explain how engineers or designers have applied the turning effect of force to make the tool more effective or efficient. Mention the pivot, the effort, and the load in each case.	Example 1 — Farming: The long-handled jembe used across Kenya's Rift Valley and highlands is a deliberate engineering choice. The metal blade is the load, the soil surface acts as the pivot, and the farmer's hands apply the effort. By designing a handle of 1–1.2 metres, tool-makers ensure the farmer can produce a large torque with moderate effort, reducing fatigue during long hours of digging. A shorter handle would require a much larger force to produce the same torque, causing rapid exhaustion. Example 2 — A wheelbarrow on a construction site in Mombasa: The wheel axle is the pivot, the load (sand or cement blocks) sits in the tray close to the wheel, and the worker lifts the handles at the far end. The long handles maximise the perpendicular distance from the pivot to the applied effort, so even a heavy load can be lifted and moved with relatively little force. Engineers position the tray close to the wheel to keep the load's moment small, meaning the worker's effort moment only needs to be slightly larger to lift the load — this is an application of both torque and the Principle of Moments working together.

Section 5: Answering the Driving Question	
Now write a full, connected answer to the driving question: 'Why does where you push, pull, or grip something determine how easy or hard it is to make it turn — and how do engineers and farmers in Kenya use this to design better tools?' Your answer should bring together all four sections above into one coherent scientific explanation. It must reference the anchoring phenomenon, the formula for torque, the Principle of Moments, and at least one Kenyan engineering application.	Where you push, pull, or grip an object determines the perpendicular distance from the pivot to the line of action of your force. According to the torque formula — Torque = Force × Perpendicular distance — a larger distance produces a greater turning effect even if your applied force stays the same. This is exactly why the farmer in Kericho finds a long-handled jembe so much easier to use: the longer handle multiplies her effort into a large torque at the blade's pivot, driving it powerfully into the soil. Gripping near the blade collapses this distance and collapses the torque, making every dig a struggle. The Principle of Moments extends this idea to balanced systems: for a tool or structure to be in equilibrium, all clockwise moments must equal all anticlockwise moments about the pivot. Kenyan engineers and farmers apply these principles deliberately in tool design. Jembe handles are made long to maximise torque with minimal effort; wheelbarrow handles are lengthened for the same reason; market balance scales use the Principle of Moments to compare unknown masses with known weights. In every case, the designer is solving the same fundamental problem the farmer in Kericho solved instinctively — that the effectiveness of a force in causing rotation depends not just on how hard you push, but on exactly where you push. Understanding torque and moments allows engineers to design smarter tools that work with human strength, not against it.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Scientific Accuracy of Torque and Moments	Torque formula is stated correctly ($\tau = F \times d$) and applied accurately in at least one worked numerical example. The Principle of Moments is defined precisely and verified mathematically. All scientific vocabulary (torque, pivot, perpendicular distance, equilibrium, clockwise/anticlockwise moment) is used correctly throughout.	Torque formula is stated and a numerical example is attempted with only minor errors. The Principle of Moments is defined correctly in words. Most scientific vocabulary is used correctly, with one or two small inaccuracies.	The torque formula is partially stated or contains errors. The Principle of Moments is mentioned but not clearly explained or verified. Scientific vocabulary is limited or frequently misused.
Explanation of the Anchoring Phenomenon	The Jembe Handle Puzzle is fully and precisely explained using torque, perpendicular distance, and pivot. The student clearly links the long handle to greater torque and the short handle to reduced torque, and explains why this changes the effort required. The explanation is coherent, causal, and scientifically accurate.	The phenomenon is explained with reference to torque and distance, and the link between handle length and ease of digging is made clear. One or two steps in the causal chain may be incomplete or slightly imprecise.	The student attempts to explain the phenomenon but the explanation is vague, relies on everyday language without scientific terms, or does not correctly connect handle length to torque.
Application to Kenyan Tools and	At least two Kenyan tools or contexts are	Two Kenyan tools are mentioned and the	Only one tool is discussed, or the tools

Engineering Contexts	discussed in detail (e.g., jembe, wheelbarrow, balance scale, door). For each, the pivot, effort, load, and turning effect are clearly identified and the engineering design decision is explained scientifically. Examples are specific and accurate.	basic application of torque is described for each. The pivot, effort, and load are identified, though the explanation of the design decision may lack full scientific depth for one example.	mentioned are not clearly connected to the science of torque and moments. The pivot, effort, and load are not clearly identified.
Use of the Principle of Moments	The Principle of Moments is stated accurately and illustrated with a clear Kenyan example. A correct numerical calculation is shown that verifies the principle (clockwise moment = anticlockwise moment). The student explains what happens to the system if the moments are unequal.	The Principle of Moments is stated correctly and a Kenyan example is given. A numerical calculation is attempted and is mostly correct, though there may be a minor arithmetic or unit error.	The Principle of Moments is mentioned but not clearly illustrated with an example. No numerical calculation is attempted, or the calculation contains significant errors that undermine the explanation.
Coherence and Quality of Final Driving Question Response	Section 5 provides a fully integrated, well-structured answer that synthesises all unit concepts (torque formula, Principle of Moments, anchoring phenomenon, engineering applications) into one connected scientific explanation. The response is written clearly, uses scientific vocabulary throughout, and directly and completely answers every part of the driving question.	Section 5 addresses the driving question and connects most major concepts from the unit. The response is mostly coherent and uses scientific vocabulary, though one concept may be underdeveloped or the link between sections could be stronger.	Section 5 attempts to answer the driving question but is incomplete, repetitive, or disconnected. Key concepts from the unit are missing or not linked together. The answer relies heavily on everyday language rather than scientific explanation.